

Ha Yun Anna Yoon

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Research Interest

Biophotonics, Systems Neuroscience, Image Processing, Machine Learning, Computer Vision

Education

University of California, Berkeley

MS/PHD MECHANICAL ENGINEERING

- Advisor: Dr. Na Ji
- Concentration: Biomechanics
- Minor: Machine Learning, Energy Science & Technology

Berkeley, CA

Expected: 2026

Massachusetts Institute of Technology

BS MECHANICAL ENGINEERING

- Advisor: Dr. Brett Bouma

Cambridge, MA

June 2019

Publications

Manuscripts

[0] Stimulation with ECoG electrodes modulates cortical activity and sensory processing in the awake mouse brain
Liang, J.F., Lee, K., Tchoe, Y., Ganji, M., Vatsyayan, R., **Yoon, H.A.**, Garrett, J., Dahye, S., Halgren, E., Ji, N. *Submitted*

[1] Effects of reducing axial resolution in two-photon calcium imaging on retrieving functional neuronal activity.
Yoon, H.A., Ferrer-Imbert, G., Charles, A., Ji, N. *In preparation*

[2] Convergence of cortical and thalamic origins of free behavior correlate in mouse V1.
Yu, P*, **Yoon, H.A.***, Yang, Y*, Xu, Y., Gozel, O., Tian, G., Ji, N., Doiron, B.
* = denotes equal contributions. *In preparation*

[3] Domain-Adaptive Poisson-Gaussian Diffusion for Low-Light Image Restoration.
Yoon, H.A., Hong, J. *In preparation*

[4] Multichannel Conditional Denoising Diffusion Models for Neuropil Decontamination.
Yoon, H.A., Hong, J. *In preparation*

Conferences

[5] Quantification of nonsense-free correlation uncovers the interaction between top-down and bottom-up sources of behavioral correlation in mouse V1.
Yu, P., **Yoon, H.A.**, Yang, Y., Xu, Y., Gozel, O., Ji, N., Doiron, B. *COSYNE 2025*.

[6] Integration of behavioral related correlation from top-down and bottom-up pathways in mouse V1.
Yu, P., **Yoon, H.A.**, Yang, Y., Gozel, O., Ji, N., Doiron, B. *COSYNE 2024*.

[7] Effects of reducing axial resolution in two-photon calcium imaging on retrieving functional neuronal activity.
Yoon, H.A., Charles, A., Ji, N. *SfN Neuroscience 2023*.

[8] Quantifying blood flow using backscattering indicator-dilution in intravascular optical coherence tomography: in vitro validation.
Uribe-Patarroyo, N., **Yoon, H.A.**, Bouma, B. *Optics in Cardiology 2018*.

[9] Flexible all-polymer multimodal fiber for integrated optogenetics.
Park, S., Guo, Y., Jia, X., Choe, H., Grena, B., Kang, J., **Yoon, H.**, Choi, G.B., Fink, Y., Anikeeva, P. *SfN Neuroscience 2016*.

Microscope Development

[10] Simulate, Design and develop 2P integrated free-space angular chirp enhanced delay (FACED) microscope v3.0.

Textbook

[0] Renewable Energy, 3rd ed.
Yoon, C.S., **Yoon, H.A.**, Yoon, J.S. *Infinity Books* 2019.

Talks

[0] Simultaneous Calcium and Voltage Imaging in mouse V1.
Samsung x Stanford KSAS x UC Berkeley KGSA Academic Conference, San Jose, CA. *Fall 2023*

Work Experience

- Tomocube, Inc.** *Daejeon, South Korea*
CLINICAL RESEARCH INTERN *Summer 2019*
- Projects: Develop deep learning techniques to identify different white blood cells in AML, APL, and Lymphoma
- Harvard Medical School- Wellman Center of Photomedicine** *Boston, MA*
ADVISORS: DR. GUILLERMO TEARNEY *2019-2020*
- Project: Develop portable, inexpensive CP-OCT system
- Harvard Medical School- Wellman Center of Photomedicine** *Boston, MA*
ADVISORS: DR. BRETT BOUMA AND DR. NESTOR URIBE-PATARROYO *2017-2019*
- Thesis: "Measuring coronary artery flow rates using intravascular optical coherence tomography to improve the assessment of percutaneous coronary intervention"
- ExxonMobil Corporation** *Spring, Texas*
UPSTREAM ENGINEERING INTERN *Summer 2018*
- Projects: Led the proposal to monitor Steel Lazy Wave Riser from Liza Phase 1 Project
 - Evaluated contract bids for Neptun Deep Project
- NASA Goddard Space Flight Center** *Greenbelt, MD*
ROBOTICS ENGINEERING INTERN *Winter 2017*
- Projects: Simulate the trajectory of robotic arms for Landsat 7 for Restore-L Mission
- Massachusetts Institute of Technology - Dept of Material Science and Engineering** *Cambridge, MA*
ADVISORS: DR. POLINA ANIKEEVA AND DR. SEONGJUN PARK *2015-2017*
- Projects: Develop flexible all-polymer multimodal fibers for integrated optogenetics
- Korea Institute of Science and Technology - Biomedical Research Institute** *Seoul, South Korea*
ADVISORS: DR. SONGJOO LEE *Summer 2016*
- Projects: Design lower limb prosthetics and Quantify balance control and lower limb postural
- Korea Institute of Machinery and Materials - Environmental and Energy Systems** *Daejeon, South Korea*
ADVISORS: DR. HONGSUK KIM *Winter 2016*
- Projects: Design in-house Selective Catalytic Reduction (SCR) pumps, and Test Three-Way-Catalytic Converter to control diesel engine emission
- Massachusetts Eye and Ear Infirmary - Eaton-Peabody Laboratory** *Boston, MA*
ADVISORS: DR. JOHN ROSOWSKI AND DR. NIMA MAFTOON *2015-2016*
- Projects: 3D CAD human middle ear using ITK-Snap and 3D Slicer to use as prosthesis

Mentoring

Brayden Ye, Undergraduate Researcher, *UC Berkeley* (2025 -)
Adele Beamer, Undergraduate Researcher, *Tufts University* (2025)
Kavish Loomba, Undergraduate Researcher, *UC Berkeley* (2024)
Trinav Chaudhuri, Undergraduate Researcher, *UC Berkeley* (2023 -)
Erin Kim, High School Researcher, *UC Berkeley* (2023)

Teaching Experience _____

UNIVERSITY OF CALIFORNIA, BERKELEY

NEU C62: Drugs and the Brain, Graduate Student Instructor, *Fall 2025*

EduExplora: Ethics, Responsibility, and Innovation in Educational AI, Instructor, *Summer 2025*

EduExplora: Medical Imaging in Clinical Diagnosis and Treatment, Instructor, *Summer 2023, 2024, 2025*

MCB Biology 1AL: General Biology Laboratory, Graduate Student Instructor, *Fall 2020, Spring 2021, Summer 2021*

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

2.001: Mechanics and Materials I, Department Tutor, *Spring 2019*

2.005: Thermal-Fluids Engineering I, Department Tutor, *Spring 2019*

2.007: Design and Manufacturing I, Department Tutor, *Spring 2019*

2.008: Design and Manufacturing II, Department Tutor, *Spring 2019*

MIT Global Teaching Lab: Girls' Town Boys' Town, Head Instructor, *Winter 2019*

Awards, Fellowships, & Grants _____

2024	Excellent Honor Scholarship – Korean Honor Scholarship , Korean Embassy in the U.S.A.	\$3000
2024-25	Graduate Diversity and Community Fellowship , UC Berkeley Office for Graduate Diversity	\$7500
2024	IEEE Photonics Society "Most Improved Chapter Award" , IEEE Photonics Society	\$200
2024	H2H8 Mentor Research Program , University of California, Berkeley	\$3,500
2023	H2H8 Fellowship , University of California, Berkeley	\$10,000
2019	MIT Martin Prince Innovation Award , Massachusetts Institute of Technology	\$3,600
2015-2019	Nordstrom Scholarship Winner , Nordstrom	\$10,000
2015	Most Valuable Student , National & Georgia Elks Association	\$11,600
2015	Regional Finalist , Coca Cola Scholars Foundation	\$1,000
2015	National Merit Scholar , National Merit Scholarship Corporation	\$2,500

Service _____

2025 - **Reviewer**, IEEE AI, Data Science, Cyber Security and Smart Manufacturing for Sustainable Development

2023-24 **SPIE, Optica, IEEE Photonics Society**, UC Berkeley Chapter President

2023 **UC Berkeley KSEA Mentor**, Graduate Mentor

2023 **UC Berkeley Society of Women Engineers Mentor**, Graduate Mentor

2021 **Faculty Search Committee - UC Berkeley MechE Energy Science & Technology**, Student Member

2020- **MIT Educational Council**, Interviewer

2017-2019 **MIT MechE Student Advisory Committee**, Committee Member

References _____

Na Ji, Professor at UC Berkeley, jina@berkeley.edu

Gary Tearney, Professor at Harvard Medical School, gtearney@partners.org

Brett Bouma, Professor at Harvard Medical School, bbouma@mit.edu

Seongjun Park, Assistant Professor at KAIST, spark19@kaist.ac.kr